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Cryoablation for Chest Wall Trauma: A Brief Report

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Abstract

Respiratory failure secondary to rib fractures is a major source of morbidity and mortality in trauma patients, particularly in older populations. Management of pain in these patients is complex due to the nature of the injuries. We present 3 patients who underwent a video-assisted thoracoscopic cryoablation of intercostal nerves for pain control after chest trauma. None of the patients developed post-operative complications related to poor respiratory status such as pneumonia or atelectasis. At one-month clinic follow-up, all patients reported no chest pain and were not using opiate analgesics. In patients for whom there is a contraindication to rib fixation in the setting of unstable rib fractures, cryoablation may be a method by which to improve respiratory status and decrease ventilator dependency due to pain. Cryoablation of intercostal nerves may provide a more durable and clinically feasible solution to aid in the healing process of these patients.

Keywords

Trauma, rib fractures, pain

Respiratory failure secondary to rib fractures is a major source of morbidity and mortality in trauma patients, particularly in older populations. Pain can decrease lung expansion and gas exchange, leading to greater ventilator time and increased risk of pulmonary complications. Management of pain in these patients is complex due to the nature of the injuries. There are a wide variety of analgesic methods available including patient-controlled pain pumps, and advanced interventions such as epidural or paravertebral catheters.[1] However, these carry limitations to their use such as epidural hematoma, hypotension, arrhythmia, and the requirement for inpatient admission with an attached device.[2] A secondary consideration is the short duration of analgesia associated with most pain control modalities. This case series seeks to describe the benefits of cryoablation as a part of multimodal analgesic strategy for patients with severe rib fractures.

We present 3 patients who underwent a video-assisted thoracoscopic cryoablation of intercostal nerves for pain control: (1) a 60-year-old male involved in MVC, with fractures of ribs 3, 7, and 9 on the left, and 3-8 on the right, with concern for osteomyelitis of vertebral body of T6; (2) a 55-year-old male involved in a boating accident 18 months prior with fractures of ribs 4-9 on the right, with chronic chest wall pain not improved by maximum outpatient therapy; and (3) a 35-year-old male involved in an MVC with fractures of ribs 1-4, 6, and 8 on right, a sternal fracture, and hemopneumothorax, readmitted with a right-sided empyema nine days after injury.

The procedure began as a standard video-assisted thoracoscopic surgery (VATS) with the patient in lateral decubitus position, and camera port inserted in the ninth intercostal rib space with a working port in the fifth intercostal rib space. Lysis of adhesions, evacuation of hemothorax, or decortication was performed as needed, after which the cryotherapy device was introduced. An Atricure CryoSphere (Mason, Oh.) probe was used as the cryotherapy device. It is programmed to reach a temperature of between minus 40 and 80 °C for a cycle of two minutes. Starting at the third rib, the tip of cryotherapy device was placed inferior to the rib approximately 2 cm

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lateral to the sympathetic chain (4 cm from the edge of the vertebral body), and the machine cycled to ablate the nerve. This process was repeated with each rib space down to the ninth rib.

All patients were extubated immediately post-operatively, except for patient 3 who was intubated prior to his procedure and was able to be extubated 2 days post-operatively. None of the patients developed post-operative complications related to poor respiratory status such as pneumonia or atelectasis. At one month clinic follow-up, all patients reported no chest pain and were not using opiate analgesics.

Pain management in chest wall trauma remains difficult due in part to the complex nature of the injuries and in part to the varied side effects of the treatment modalities available. In an effort to prevent hypoventilation in this patient population, pain management techniques such as patient-controlled analgesic pumps, neuro-axial blockade, and intercostal nerve blockade may be utilized, but are limited in duration of relief provided, need for device access, and side effect profiles.

Thoracoscopic surgery in trauma patients is frequently undertaken for several indications such as empyema, retained hemothorax, recurrent pneumothoraces, and, most recently, by some institutions for rib fixation. In these instances, cryoablation should be considered as a long-term pain modality, particularly those with rib fractures.[3,4] In patients for whom there is a contraindication to rib fixation in the setting of unstable rib fractures, cryoablation may be a method by which to improve respiratory status and decrease ventilator dependency due to pain. Cryoablation of intercostal nerves may provide a more durable and clinically feasible solution to aid in the healing process of these patients.

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